Exercise Sheet 01

The Notice Board

House rules.

- One file per exercise, in `my_work/`: `ex1_loot_maths.py`, `ex2_names.py`, …
- Start each file by typing the **Given** lines. Then write the rest yourself.
- Run it (`python my_work/ex1_loot_maths.py`) and compare with the **expected output**, character by character.
- Stuck for more than 10 minutes? Look at the cheat sheet, then skip and come back later.
- Hand in: commit and push to `lesson-01`, then open a pull request. Tell Lars which raven was the most annoying.

Exercise 1 — Loot maths

10 min · `ex1_loot_maths.py`

> **RAVEN #101, from Pell, the Quartermistress:** *"The Grimoire failed again at the full moon. Run-time error '13', naturally. Can you redo the basic loot maths in Python? Yours in haste, P."*

GIVEN

```
reward = 1000
party_size = 3
quest_days = 17
supplies_per_day = 12.5
bounty = 2600
growth_percent = 10
months = 3
```

Calculate and print, rounded to 2 decimal places where it makes sense:

(a) each hero's share of the reward;

(b) the share in whole crowns, and how many crowns are left over for the guild chest (use `//` and `%`);

(c) how many full weeks and leftover days the quest takes;

(d) the cost of supplies for the whole quest;

(e) the bounty on the Wyrm after `months` months, if it grows by `growth_percent` % every month, each on top of the last.

EXPECTED OUTPUT

```
Share per hero: 333.33
Whole crowns each: 333 and left for the guild chest: 1
Quest length: 2 weeks and 3 days
Supplies: 212.5
Wyrm bounty after 3 months: 3460.6
```

Hint: "To the power of" is `**` in Python. If you use `^`, you'll meet it again in the Grimoire.

Exercise 2 — Names and text

10 min · `ex2_names.py`

> **RAVEN #102, from Orrin of the Clockworks:** *"The Scriptorium's copy of the notice board is a mess. Dalvric's incantation 'cleaned' it by burning every second notice. Please be gentler than Dalvric."*

GIVEN

```
raw_first = "  wREN "
raw_last = "hawkRIDGE  "
raw_id = "  q-0042 "
```

(a) Clean the names (no spaces around them, proper capitals) and print the full name.

(b) Print the initials with dots.

(c) Print the raven address: *first.last*@thornmere.realm, all lowercase.

(d) Clean the quest ID: no spaces, uppercase, no dash.

(e) Captain Siv's squire is called `"Mary Kate"` `"Oldcastle"`. Print her raven address, without the space.

EXPECTED OUTPUT

```
Wren Hawkridge
W.H.
wren.hawkridge@thornmere.realm
Q0042
marykate.oldcastle@thornmere.realm
```

Exercise 3 — A notice the Queen can love

10 min · `ex3_notice.py`

> **Decree from Queen Aldara, sent at dusk:** *"The notices on the board look 'a bit 1019'. Commas! Things lined up! Enjoy the rest day. A."*

GIVEN

```
quest = "Drive off trolls at Larkspur pass"
region = "Ironhills"
reward = 4860
party_size = 4
cut_rate = 0.10
supplies = 96.40
```

The guild takes its cut (10 % of the reward). What's left after the cut and the supplies is shared equally by the party. Print exactly this:

EXPECTED OUTPUT

```
NOTICE: Drive off trolls at Larkspur pass
---------------------------
Reward          4,860.00 cr
Guild cut         486.00 cr
Supplies           96.40 cr
---------------------------
Per hero        1,069.40 cr

Guild cut: 10.0%
Region: Ironhills, party of 4
```

Hint: Each money line is 27 characters: the label left-aligned in 15, the amount right-aligned in 12. Make the money text first: `reward_text = f"{reward:,.2f} cr"`, then `f"{'Reward':<15}{reward_text:>12}"`. And `"-" *`

`27` draws the line.

Exercise 4 Text that should be numbers

10 min · `ex4_conversions.py`

> **RAVEN #104, from Pell:** *"The Scriptorium copies EVERYTHING out as text. Apparently '4,860.00 cr' is not a number. The Grimoire says 'Type mismatch'. I say 'please'."*

GIVEN

```
reward_text = "4,860.00 cr"
days_text = " 9 "
cut_text = "12.5%"
open_text = " Yes "
quest_number = 42
```

(a) Turn the reward into a decimal number, the days into a whole number, the cut into a fraction (0.125) and `open_text` into `True` / `False`. Print each value together with its `type()`.

(b) Build the quest ID: `Q` followed by the number padded to 4 digits.

(c) Print the guild's cut in crowns (reward × cut).

EXPECTED OUTPUT

```
4860.0 <class 'float'>
9 <class 'int'>
0.125 <class 'float'>
True <class 'bool'>
Q0042
Guild cut: 607.50 cr
```

Exercise 5 Yes or no

5 min · `ex5_true_false.py`

> **RAVEN #105, from Captain Siv, quest master:** *"Just yes or no, please. Dalvric's version once answered 'mayhap'."*

GIVEN

```
hero_level = 7
party_size = 3
difficulty = 4
reward = 1300
min_reward = 400
max_reward = 1200
```

(a) Print four answers. **Veteran**: level 10 or more. **Party may go**: at least 3 heroes *and* difficulty 4 or less. **Normal reward**: reward between min and max, limits included. **Needs the Queen's seal**: not a normal reward.

(b) Before running it again: if `hero_level = 12` and `reward = 1200`, what will it print? Write your guess as a comment at the bottom of the file, then try it.

EXPECTED OUTPUT

```
Veteran: False
Party may go: True
Normal reward: False
Needs the Queen's seal: True
```

Exercise 6 LEFT, RIGHT, MID

10 min · `ex6_slicing.py`

> **RAVEN #106, from Orrin:** *"In Excel I'd use LEFT/MID/RIGHT. Dalvric used four hundred lines. Please use fewer."*

GIVEN

```
realm = "Thornmere"
ref = "GOBLINS-1203"
```

(a) Print the Python versions of `LEFT(realm, 5)`, `RIGHT(realm, 4)` and `MID(realm, 3, 4)`.

(b) Split `ref` at the dash: print the monster, then the number as a real number.

(c) Print the number plus one, to prove it really is a number now.

EXPECTED OUTPUT

```
Thorn
mere
ornm
GOBLINS
1203
1204
```

Hint: Excel counts from 1, Python from 0. `ref.find("-")` tells you where the dash is. Your code should still work if the ref were `"WYRM-0007"`. Try it!

Exercise 7 The guild in one sentence STRETCH

15 min · `ex7_one_sentence.py`

> **Decree from Queen Aldara:** *"The Privy Council wants the guild's record 'in one sentence a queen can say out loud'. Council sits Thursday. No pressure. (Some pressure.) P.S. Biscuit's flight log is in decimal hours and nobody understands it. A."*

GIVEN

```
quests = 400
successes = 259
flight_hours = 1.75
```

The success rate is **successes / quests × 100**, rounded to 1 decimal. "Quests won out of every 100" is the same thing rounded to a whole number.

EXPECTED OUTPUT

```
Our heroes win 65 of every 100 quests they take (success rate: 64.8%).
Biscuit flew for 1h 45m.
```

Hint: `int(1.75)` is `1`. The minutes come from what's left over. `{minutes:02d}` pads to two digits.

Bug hunt Pages from the Grimoire, page 1

15 min · `bughunt/loot_bug.py`

Dalvric translated a corner of the Grimoire into Python and sailed west. Since then the Wyrm bounty crashes, and no hero has paid the guild its cut for three months. **Three** bugs: one crashes, one is silently wrong, one you'll only spot letter by letter.

EXPECTED OUTPUT ONCE FIXED

```
Wyrm bounty after 2 months: 2,205.00 cr
Osric Kettleby: loot 2345.50 cr, guild cut 469.10 cr, keeps 1876.40 cr
```

Commit each fix separately, explained like a detective's report.

More riddles. Five more torn pages wait in `bughunt/riddles/`, one bug each (the last one has two), getting harder as you go. Optional: for fast finishers, or a quiet evening.